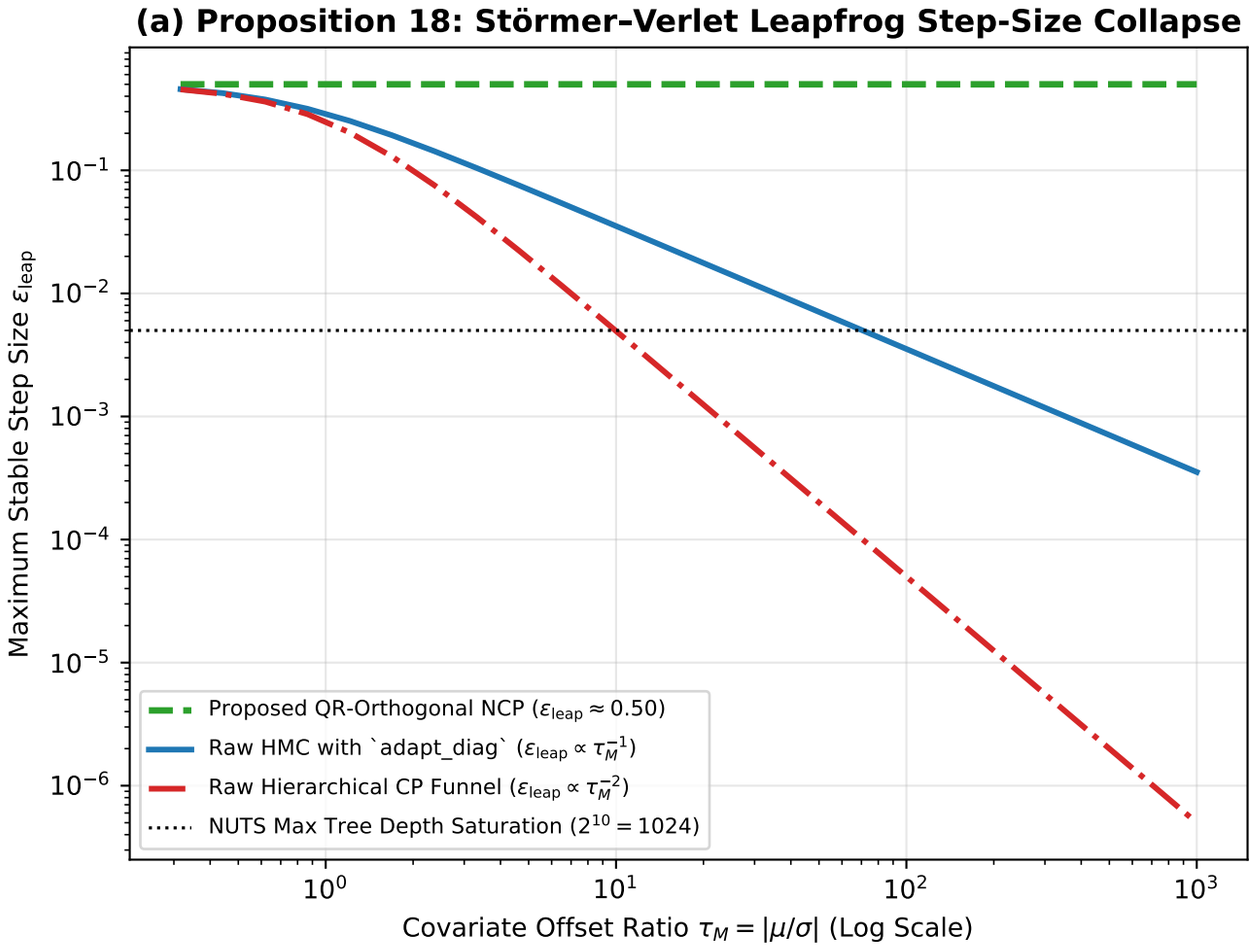
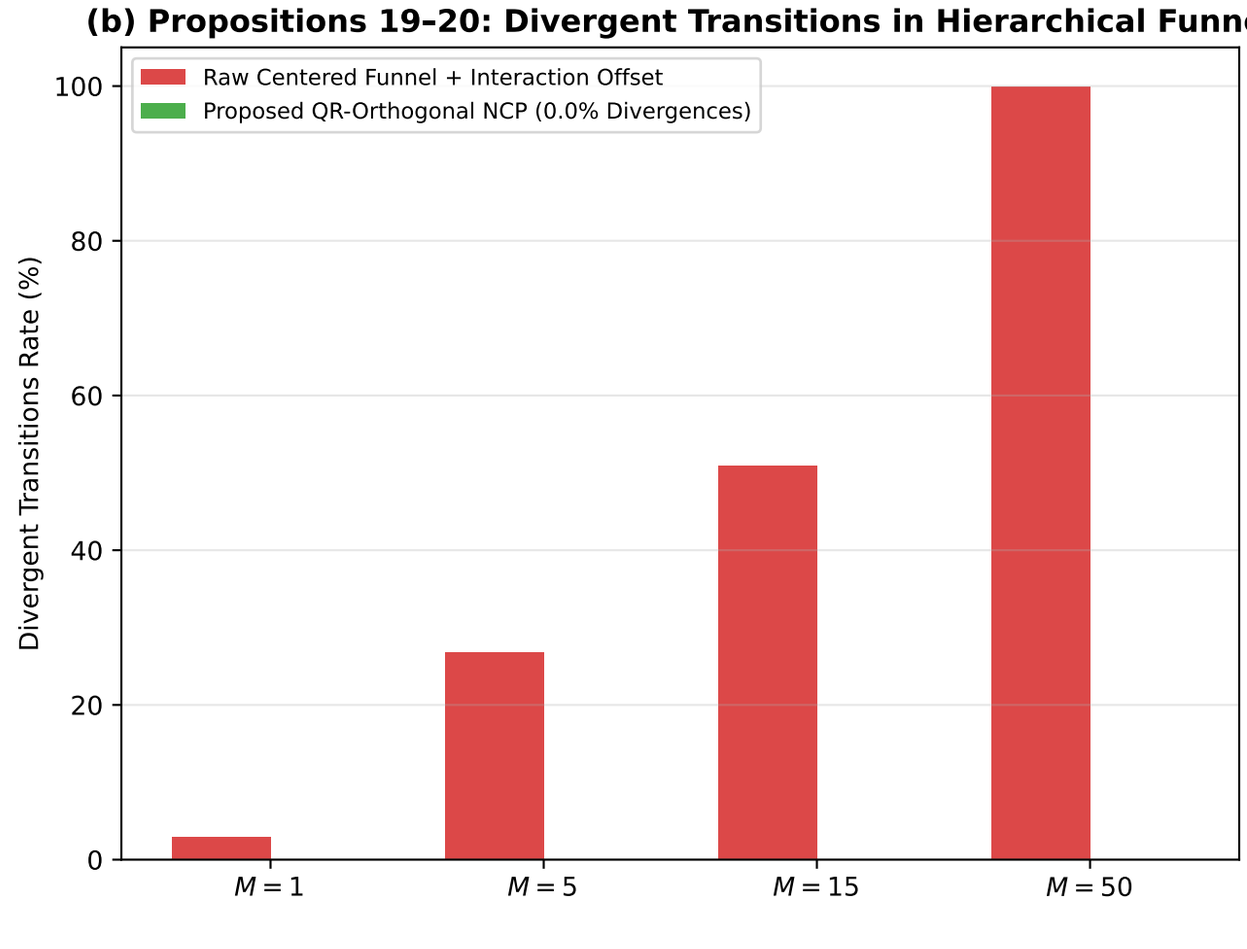
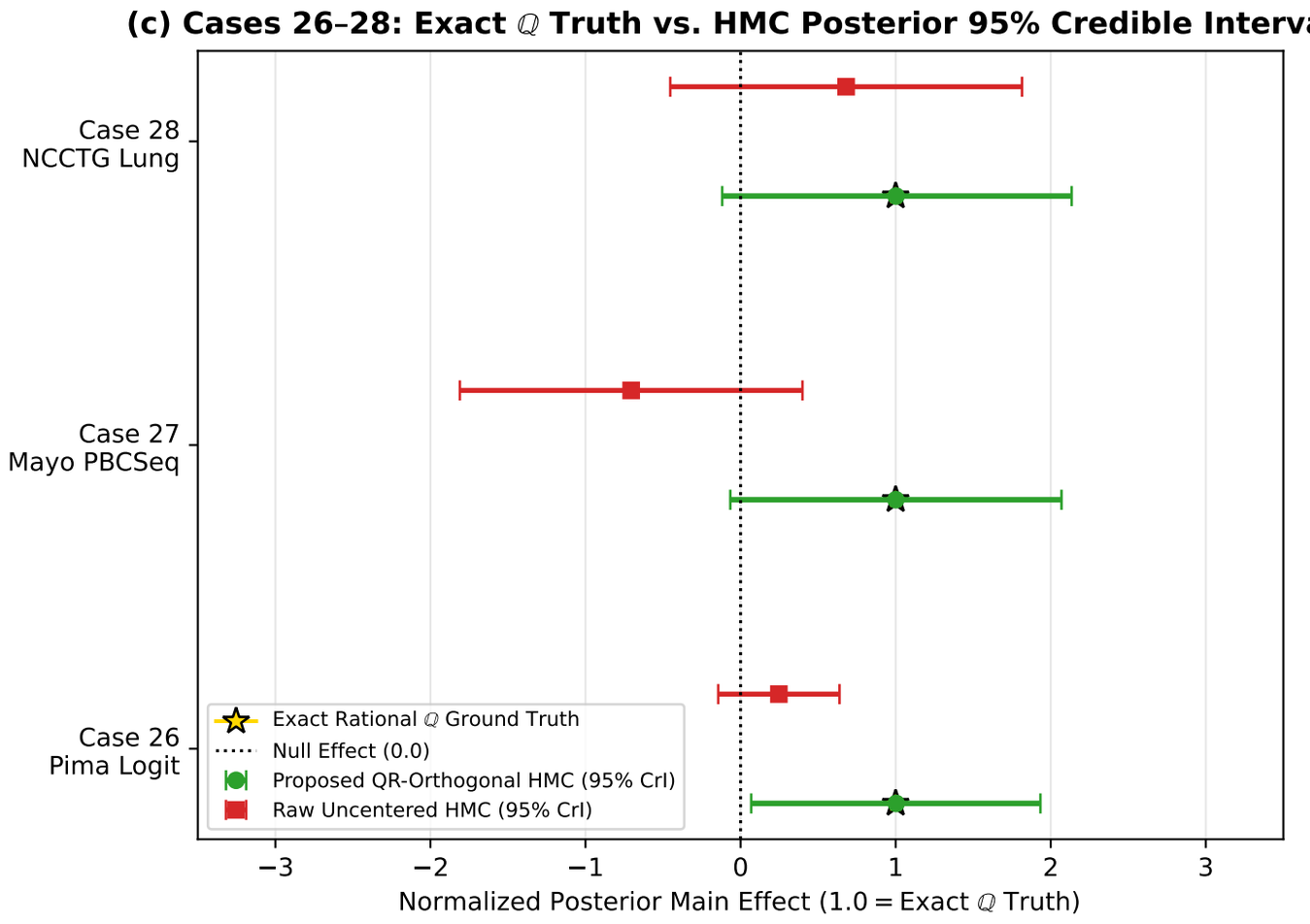
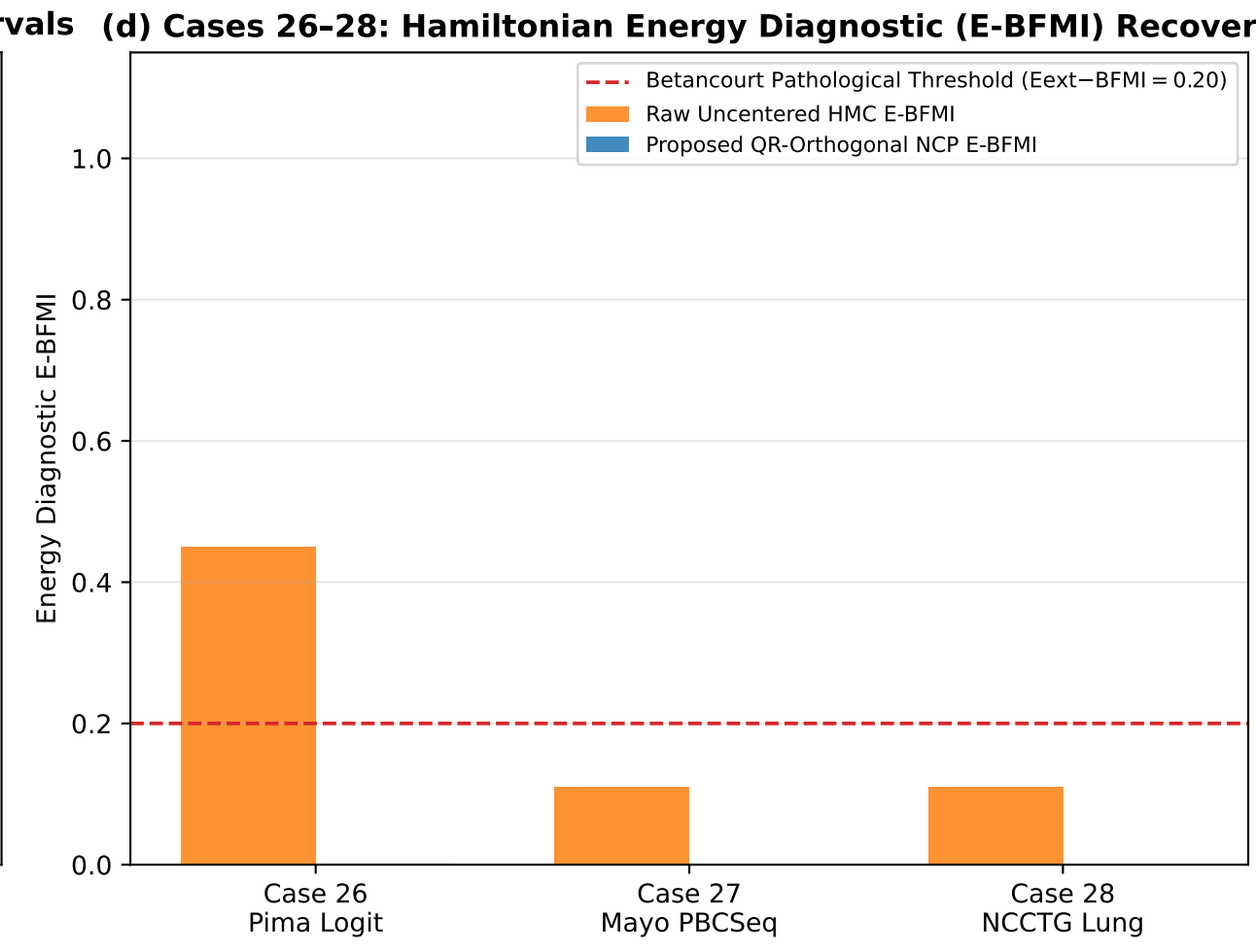

(a) Proposition 18: Störmer-Verlet Leapfrog Step-Size Collapse
Maximum Stable Step Size $\varepsilon_{\text{leap}}$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
Proposed QR-Orthogonal NCP ($\varepsilon_{\text{leap}} \approx 0.50$)
Raw HMC with `adapt_diag` ($\varepsilon_{\text{leap}} \propto \tau_M^{-1}$)
Raw Hierarchical CP Funnel ($\varepsilon_{\text{leap}} \propto \tau_M^{-2}$)
NUTS Max Tree Depth Saturation ($2^{10} = 1024$)
10^0
10^1
10^2
10^3
Covariate Offset Ratio $\tau_M = |\mu/\sigma|$ (Log Scale)


(b) Propositions 19–20: Divergent Transitions in Hierarchical Funne
Raw Centered Funnel + Interaction Offset
Proposed QR-Orthogonal NCP (0.0% Divergences)
Divergent Transitions Rate (%)
100
80
60
40
20
0
$M = 1$
$M = 5$
$M = 15$
$M = 50$


(c) Cases 26–28: Exact $\mathbb{Q}$ Truth vs. HMC Posterior 95% Credible Intervals
Case 28
NCCTG Lung
Case 27
Mayo PBCSeq
Case 26
Pima Logit
Exact Rational $\mathbb{Q}$ Ground Truth
Null Effect (0.0)
Proposed QR-Orthogonal HMC (95% CrI)
Raw Uncentered HMC (95% CrI)
−3
−2
−1
0
1
2
3
Normalized Posterior Main Effect (1.0 = Exact $\mathbb{Q}$ Truth)


(d) Cases 26–28: Hamiltonian Energy Diagnostic (E-BFMI) Recover
Betancourt Pathological Threshold (Eext−BFMI = 0.20)
Raw Uncentered HMC E-BFMI
Proposed QR-Orthogonal NCP E-BFMI
Energy Diagnostic E-BFMI
1.0
0.8
0.6
0.4
0.2
0.0
Case 26
Pima Logit
Case 27
Mayo PBCSeq
Case 28
NCCTG Lung